Cambridge Assessment Admissions Testing

NAZARBAYEV UNIVERSITY

CENTER FOR PREPARATORY STUDIES

Nazarbayev University Entrance Test

MATHEMATICS

SAMPLE QUESTIONS WITH SOLUTIONS

for

NUET SET 2022

v 1.0 January 2022

Mathematics

Question 1

In statistics, Spearman's rank correlation coefficient is given by the formula:

$$r = 1 - \frac{6\sum d^2}{n\left(n^2 - 1\right)}$$

Rearrange the formula to make $\sum d^2$ the subject.

A $\sum d^2 = 1 - \frac{r(n^3 - n)}{6}$

B $\sum d^2 = \left(\frac{(1 - r)(n^3 - n)}{6}\right)^2$

C $\sum d^2 = \frac{(1 - r)(n^3 - 1)}{6}$

D $\sum d^2 = \frac{(1 + r)(n^3 - n)}{6}$

E $\sum d^2 = \frac{(1 - r)(n^3 - n)}{6}$

Explanation 1

To rearrange a formula, the 'subject' has to be isolated.

In the formula $r = 1 - \frac{6\sum d^2}{n(n-1)}$ the $\sum d^2$ term has to be isolated.

There are several different ways of doing this but these all include the same steps.

In the first instance add $\frac{6\sum d^2}{n(n-1)}$ to both sides of the equation.

This gives $r + \frac{6\sum d^2}{n(n-1)} = 1$

Then subtract r from both sides of the equation.

This gives $\frac{6\sum d^2}{n(n^2-1)} = 1 - r$

Now divide both sides of the equation by 6 to give $\frac{\sum d^2}{n(n^2-1)} = \frac{1-r}{6}$

Finally multiply both sides of the equation by $n(n^2-1)$ to give $\sum d^2 = \frac{1-r}{6}(n)(n^2-1)$

This is not exactly the same as any of the given answers but $n(n^2-1) = n^3 - n$

So the answer arrived at is equivalent to response E.

Therefore the correct answer is **E**: $\sum d^2 = \frac{(1-r)}{6}(n^3 - n)$

Question 2

A solid sphere of radius r fits inside a hollow cylinder. The cylinder has the same internal diameter and length as the diameter of the sphere.

The volume of a sphere is $\frac{4}{3}\pi r^3$, where r is the radius of the sphere.

What fraction of the space inside the cylinder is taken up by the sphere?

A $\frac{1}{4}$

B $\frac{1}{3}$

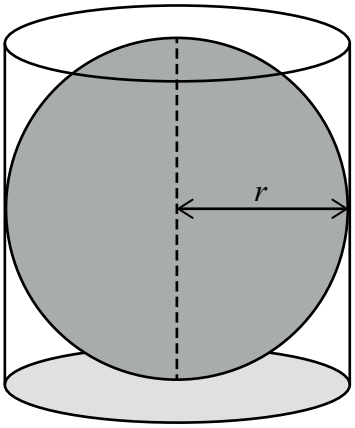
C $\frac{1}{2}$

D $\frac{2}{3}$

E $\frac{3}{4}$

Explanation 2

The volume of a cylinder (a circular prism) is the base area times the height, $\pi r^2 h$.



As the cylinder's height and internal diameter are the same as the diameter of the sphere, both of these are equal to $2r$.

This means that the volume of the cylinder is $\pi r^2 \times 2r$

The volume of the cylinder is $\pi r^2 \times 2r = 2\pi r^3$

The volume of the sphere is $\frac{4}{3}\pi r^3$

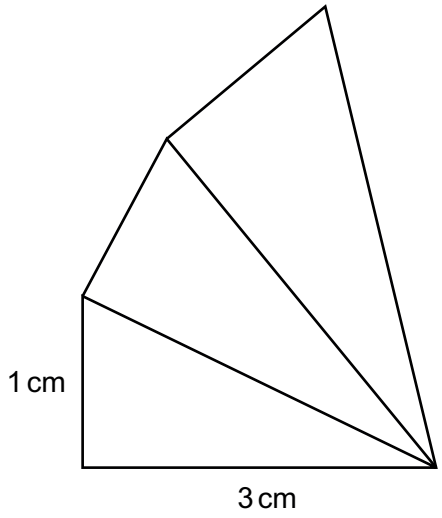
The fraction of space inside the cylinder taken up by the sphere is $\frac{\frac{4}{3}\pi r^3}{2\pi r^3}$

Simplify by cancelling the πr^3 terms to leave $\frac{4}{3} \times \frac{1}{2} = \frac{2}{3}$

Therefore the correct answer is **D,** $\frac{2}{3}$

Question 3

The diagram shows three similar right-angled triangles.



[diagram not to scale]

What is the area of the largest triangle, in cm^2?

A $\frac{5}{3}$

B $\frac{50}{27}$

C 5

D 15

E $\frac{50}{3}$

Explanation 3

Using Pythagoras' Theorem, we can find that the length of the hypotenuse in the lowest triangle of the spiral is $\sqrt{1^2 + 3^2} = \sqrt{10}$

As the triangles are similar, the scale factor of the enlargement is $\frac{\sqrt{10}}{3}$

This means that the hypotenuse in the second triangle is of length $\sqrt{10} \times \frac{\sqrt{10}}{3} = \frac{10}{3}$

Using the similarity, the other short side in the third triangle is $\frac{1}{3}$ of this, which equals $\frac{10}{9}$

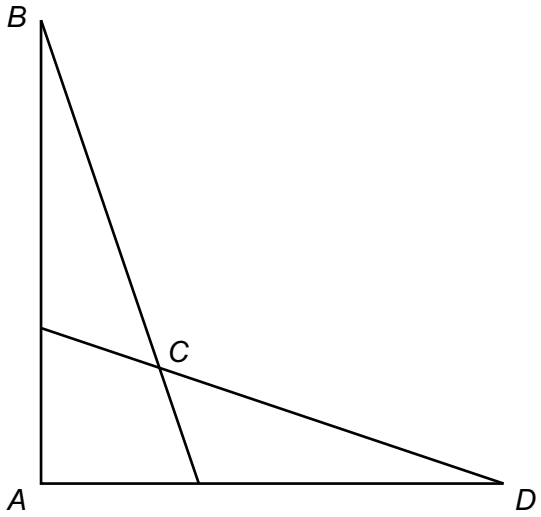
Area of triangle $= \frac{1}{2} \times$ base $\times$ height

$$= \frac{1}{2} \times \frac{10}{3} \times \frac{10}{9} = \frac{50}{27}$$

The correct response is therefore **B**.

Question 4

Two identical copies of a right-angled triangle are placed on top of each other to make a quadrilateral *ABCD*, as shown below.



[diagram not to scale]

The perpendicular sides of the triangles are in the ratio 1 : 3

What fraction of the area of the quadrilateral is covered by both triangles?

A $\frac{1}{2}$

B $\frac{1}{3}$

C $\frac{1}{4}$

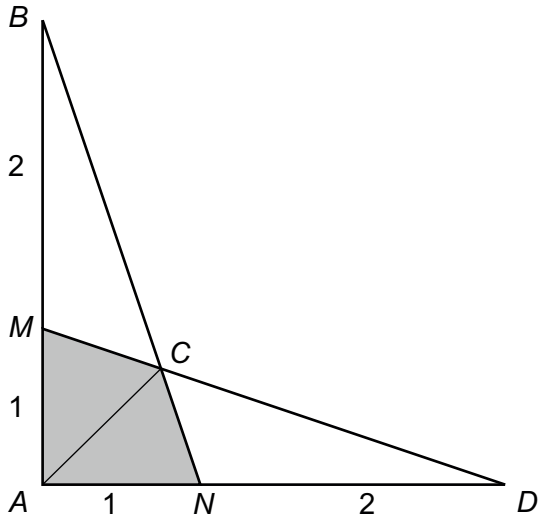
D $\frac{1}{5}$

E $\frac{1}{6}$

F $\frac{1}{7}$

Explanation 4

The area covered by both triangles is shown shaded in the diagram.



As the perpendicular sides are in the ratio 1 : 3, we can let the sides be of length 1 unit and 3 units which gives the lengths shown on the diagram.

The shaded area is made up of triangles *CAN* and *ACM*.

Triangles *CAN* and *DCN* both have the same height. The base of *DCN*, *DN*, is twice as long as the base of triangle *CAN*, *AN*, and so the triangle *DCN* has twice the area of *CAN*.

Let the area of triangle *CAN* be x square units. The area of triangle *DCN* is $2x$ square units.

The same reasoning applies to triangles *CAN* and *BCM*.

The shaded area is $2x$ square units.

The area of quadrilateral *BADC* is $2x + x + x + 2x = 6x$

The fraction covered by both triangles is $\frac{2x}{6x} = \frac{1}{3}$

The correct response is therefore **B**.

Question 5

For what values of the non-zero real number a does the quadratic equation

$$ax^2 + (a - 2)x = 2$$

have real distinct roots?

A All values of a

B $a = -2$

C $a > -2$

D $a \neq -2$

Explanation 5

A quadratic $ax^2 + bx + c = 0$ has real distinct roots when the discriminant is greater than zero; in other words, when

$$\sqrt{b^2 - 4ac} > 0$$

Recall this condition comes from considering the general formula for the solution of quadratic equations:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

In this case, for real distinct roots this condition gives:

$$(a - 2)^2 - 4a(-2) > 0$$

This simplifies to:

$$a^2 + 4a + 4 > 0$$

$$(a + 2)^2 > 0$$

This last inequality is satisfied for all values of a except $a = -2$

The correct answer is therefore **D**, $a \neq -2$

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